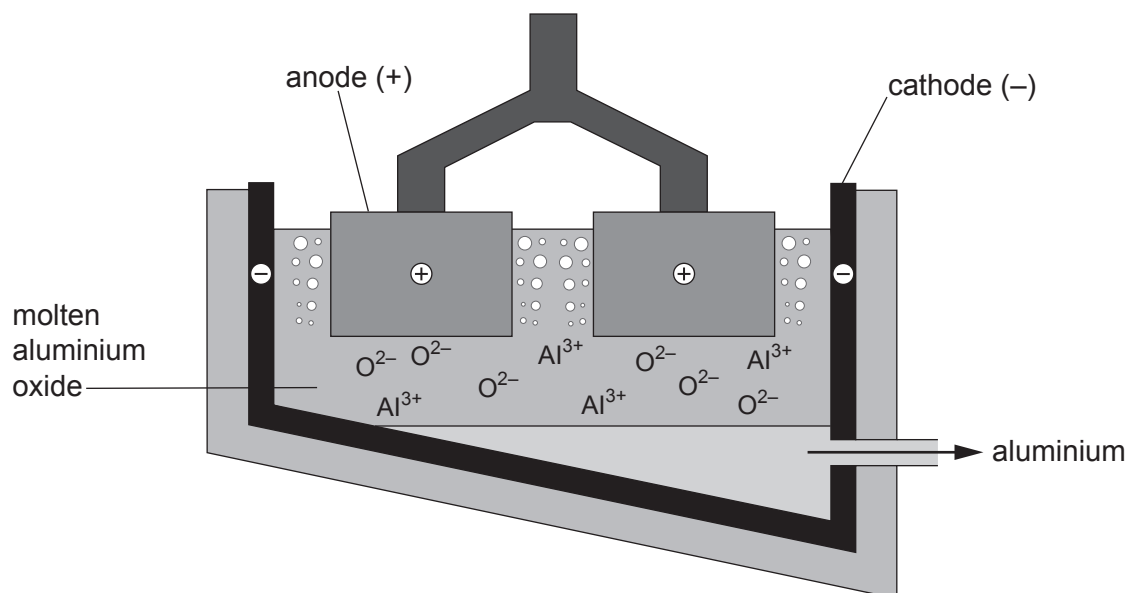


9. (a) The diagram shows a model of the cell used in the extraction of aluminium from molten aluminium oxide.



The overall equation for the reaction is shown below.



Aluminium oxide contains  $\text{Al}^{3+}$  and  $\text{O}^{2-}$  ions.

Explain, in terms of these ions, the formation of both products. Include electrode equations to support your answer. [6 QER]

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- (b) Apart from a workforce and local transport infrastructure, explain **one** other important factor when choosing a site for an aluminium extraction plant in the UK. [2]

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